

GVPT622 Lab Session

Relationships Between Two Variables

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library(tidyverse)
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library(modelsummary)

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```

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library(tidyverse)
library(broom)
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cel <- read_excel("CELHouse93to118.xlsx", guess_max = 20000)
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- ▶ `broom` and `modelsummary` are new: install them with `install.packages()` if you haven't already

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- ▶ To answer, we need an **outcome** (dependent) variable and one or more **explanatory** (independent) variables that might determine it
- ▶ Today: how do we describe the relationship between *two* variables?
 - ▷ Do they move together? In which direction? How strongly?
 - ▷ If one changes by some amount, how much do we expect the other to change?

Watch the demonstration here: [Click to Watch Video](#)

Bivariate Relationships

Case Study: What Makes a Lawmaker Effective?

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Case Study: What Makes a Lawmaker Effective?

- ▶ The CEL's **Legislative Effectiveness Score** (LES) measures how successfully each member moves their own bills through the lawmaking process
 - ▷ Credit for each bill sponsored, more for each stage it clears (committee, floor, law), and more for substantive and significant bills
 - ▷ Scaled so the average member in each Congress scores 1
- ▶ Our question: **do members who sponsor more bills have higher effectiveness scores?**
- ▶ Caveat: sponsoring is one piece of the LES, so some relationship is built in. What we want to know is *how much*: does sponsoring more translate into moving more through the process?

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```
house118 <- cel |>
  rename(
    congress = "Congress number",
    name = "Legislator name, as given in THOMAS",
    bills = "Total number of bills sponsored by this member...73",
    les = "LES 2.0",
    chair = "1 = committee chair, according to Almanac of American
    Politics",
    democrat = "1 = Democrat"
  ) |>
  filter(congress == 118) |>
  select(name, bills, les, chair, democrat)
```

Who Sponsored Nothing?

- ▶ Both variables are heavily right-skewed, so we'll want to log them, but $\log(0)$ is undefined. Who has a zero?

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```
house118 |>  
  filter(bills == 0)
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```
house118 |>  
  filter(bills == 0)
```

	name	bills	les	chair	democrat
1	Jeffries, Hakeem	0	0	0	1
2	Pelosi, Nancy	0	0	0	1

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```
house118 |>  
  filter(bills == 0)
```

	name	bills	les	chair	democrat
1	Jeffries, Hakeem	0	0	0	1
2	Pelosi, Nancy	0	0	0	1

- ▶ Top party leaders typically don't sponsor bills, so these two aren't typical members with a score of zero. They're a different kind of member

- ▶ We drop them and log both variables:

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```
house118 <- house118 |>
  filter(bills > 0) |>
  mutate(log_bills = log(bills),
         log_les = log(les))
```

Dropping Zeros and Logging

- ▶ We drop them and log both variables:

```
house118 <- house118 |>
  filter(bills > 0) |>
  mutate(log_bills = log(bills),
         log_les = log(les))
```

- ▶ That leaves $n = 446$ members of the House. Always know how many observations you dropped, and why

Visualizing the Relationship

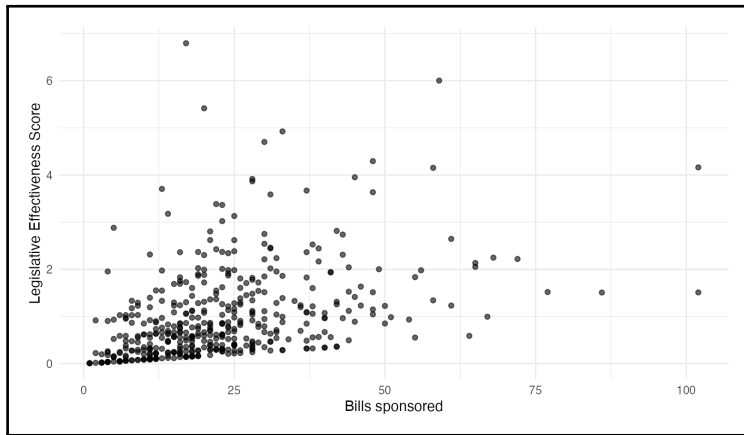
```
ggplot(house118, aes(x = bills, y = les)) +  
  geom_point(alpha = 0.6) +  
  labs(x = "Bills sponsored", y = "Legislative Effectiveness Score") +  
  theme_minimal()
```

Visualizing the Relationship

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ggplot(house118, aes(x = bills, y = les)) +  
  geom_point(alpha = 0.6) +  
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  theme_minimal()
```

- Each point is one member of the 118th House ($n = 446$)

Visualizing the Relationship



More bills tends to go with a higher score, but most members are bunched in the bottom-left corner and a handful of heavy sponsors stretch the x-axis to 102.

Logging Both Variables

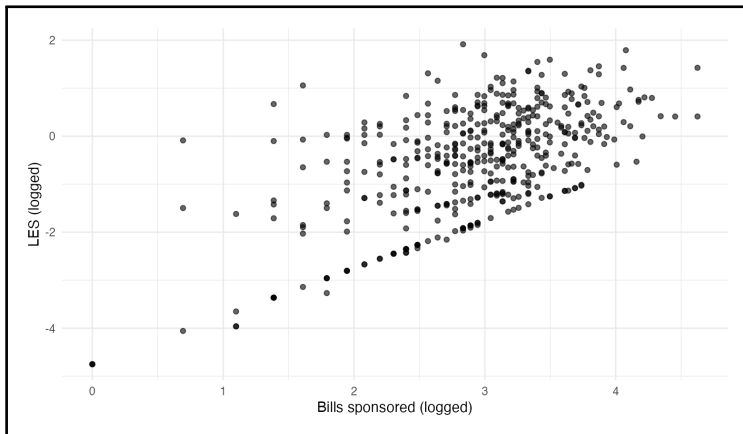
```
ggplot(house118, aes(x = log_bills, y = log_les)) +  
  geom_point(alpha = 0.6) +  
  labs(x = "Bills sponsored (logged)", y = "LES (logged)") +  
  theme_minimal()
```

Logging Both Variables

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```

- ▶ Logging pulls in the long right tails, so the relationship is spread out where we can see it

Logging Both Variables



Now the points form a clearer, roughly linear cloud. The sharp diagonal edge along the bottom is a *floor* built into the LES: every bill introduced earns a little credit, so members whose bills went nowhere line up along it.

Measuring the Strength of a Relationship

- ▶ The **correlation coefficient** (r) summarizes a linear relationship in a single number between -1 and 1

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```
cor(house118$bills, house118$les)
cor(house118$log_bills, house118$log_les)
```

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 - ▷ Sign: direction (positive or negative)
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```
cor(house118$bill, house118$les)
cor(house118$log_bill, house118$log_les)
```

```
[1] 0.409567
[1] 0.6027172
```

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cor(house118$log_bills, house118$log_les)
```

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[1] 0.409567
[1] 0.6027172
```

- ▶ Moderate and positive on the raw scale; **strong and positive** once logged. Correlation measures *linear* association, and the logged relationship is closer to a line

Building a Generalizable Description

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$$y = a + mx$$

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- ▶ For that, we summarize the cloud of points with a line:

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- ▶ **Ordinary least squares (OLS)** regression draws the line that minimizes the (squared) distance between itself and all the data points

Adding the OLS Line

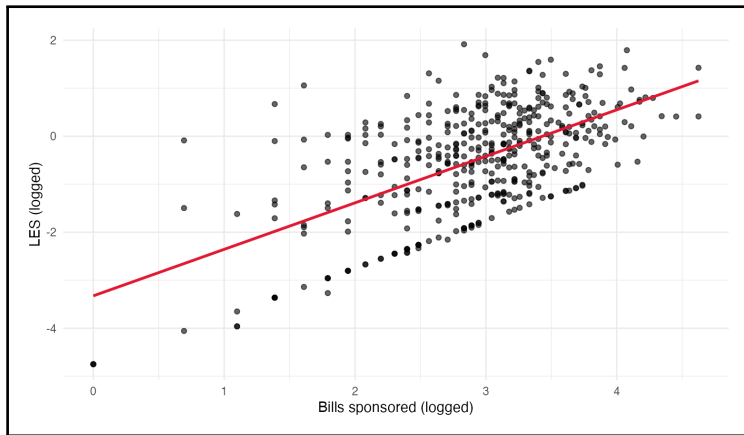
```
ggplot(house118, aes(x = log_bills, y = log_les)) +  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "#E21833") +  
  labs(x = "Bills sponsored (logged)", y = "LES (logged)") +  
  theme_minimal()
```

Adding the OLS Line

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ggplot(house118, aes(x = log_bills, y = log_les)) +  
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```

- ▶ `method = "lm"` asks for a straight OLS line; `se = FALSE` hides the uncertainty band (for now)

Adding the OLS Line



The line of best fit through the logged data.

- ▶ The model we're fitting:

$$\log(\text{LES}_i) = \beta_0 + \beta_1 \log(\text{Bills}_i) + \epsilon_i$$

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m <- lm(log_les ~ log_bills, data = house118)
m
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m <- lm(log_les ~ log_bills, data = house118)
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```
Coefficients:
(Intercept)    log_bills
   -3.3254         0.9687
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- ▶ $\beta_0 = -3.33$: the predicted logged LES when logged bills is 0 (i.e., one bill sponsored)
- ▶ $\beta_1 = 0.97$: a one-unit increase in logged bills is associated with a 0.97 increase in logged LES

Where Do the Coefficients Come From?

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

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```
house118 |>
  summarise(
    beta_1 = sum((log_bills - mean(log_bills)) *
                  (log_les - mean(log_les))) /
              sum((log_bills - mean(log_bills))^2),
    beta_0 = mean(log_les) - beta_1 * mean(log_bills)
  )
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beta_1 beta_0
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  )
```

```
beta_1 beta_0
0.969   -3.33
```

- Computed by hand: the same answer `lm()` gave. The line always passes through $(\bar{x}, \bar{y})$

A Presentable Table

```
modelsummary(m,  
  statistic = NULL,  
  coef_rename = c("log_bills" = "Bills sponsored (logged)"),  
  gof_map = "nobs")
```

A Presentable Table

```
modelsummary(m,  
  statistic = NULL,  
  coef_rename = c("log_bills" = "Bills sponsored (logged)"),  
  gof_map = "nobs")
```

	(1)
(Intercept)	-3.325
Bills sponsored (logged)	0.969
Num.Obs.	446

Interpreting a Log-Log Slope¹

- ▶ “A one-unit increase in logged bills” is hard to picture. When *both* sides are logged, the slope has a friendlier reading:

¹Check this joke out

Interpreting a Log-Log Slope¹

- ▶ “A one-unit increase in logged bills” is hard to picture. When *both* sides are logged, the slope has a friendlier reading:
- ▶ A 1% increase in bills sponsored is associated with about a 0.97% increase in LES

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- ▶ “A one-unit increase in logged bills” is hard to picture. When *both* sides are logged, the slope has a friendlier reading:
- ▶ A 1% increase in bills sponsored is associated with about a 0.97% increase in LES
- ▶ Put another way: a member who sponsors twice as many bills is predicted to have roughly twice the effectiveness score

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Interpreting a Log-Log Slope¹

- ▶ “A one-unit increase in logged bills” is hard to picture. When *both* sides are logged, the slope has a friendlier reading:
- ▶ A 1% increase in bills sponsored is associated with about a 0.97% increase in LES
- ▶ Put another way: a member who sponsors twice as many bills is predicted to have roughly twice the effectiveness score
- ▶ Sponsoring more doesn't mean each bill does worse. On average, effectiveness keeps pace almost one-for-one

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```
m_res <- tidy(m)
beta_0 <- m_res |> filter(term == "(Intercept)") |> pull(estimate)
beta_1 <- m_res |> filter(term == "log_bills") |> pull(estimate)

log_les_40 <- beta_0 + beta_1 * log(40)
log_les_40
exp(log_les_40)
```

- What LES do we predict for a member who sponsors 40 bills (roughly double the median of 20)?

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```

```
[1] 0.2478607
[1] 1.281281
```

Making Predictions

- ▶ What LES do we predict for a member who sponsors 40 bills (roughly double the median of 20)?

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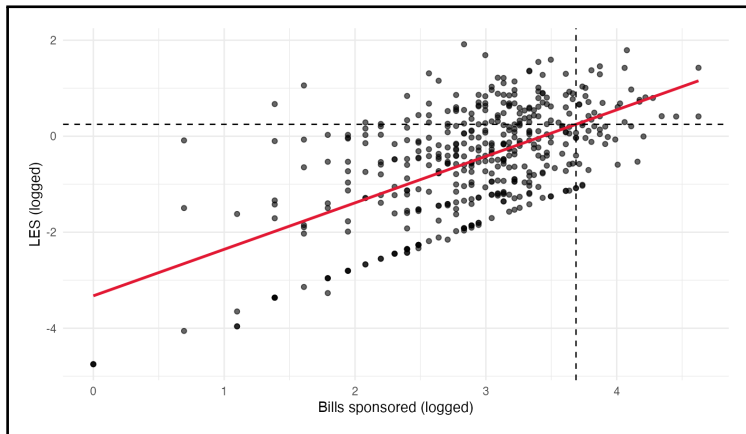
log_les_40 <- beta_0 + beta_1 * log(40)
log_les_40
exp(log_les_40)
```

```
[1] 0.2478607
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```

- ▶ `exp()` undoes the log: a predicted LES of about 1.28, a bit above the average member's score of 1

```
ggplot(house118, aes(x = log_bills, y = log_les)) +  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "#E21833") +  
  geom_vline(xintercept = log(40), linetype = "dashed") +  
  geom_hline(yintercept = log_les_40, linetype = "dashed") +  
  labs(x = "Bills sponsored (logged)", y = "LES (logged)") +  
  theme_minimal()
```

Making Predictions



The prediction sits on the line where the dashed lines cross. But look at the actual members near $\log(40) \approx 3.7$: their scores are scattered well above and below it. The model is a generalization, not a description of any one member.

Measuring Error in the Model

- ▶ The **residual** is the gap between what we observed and what the model predicted:

$$y_i - \hat{y}_i$$

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```
augment(m) |>  
  select(log_les, .fitted, .resid)
```

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$$y_i - \hat{y}_i$$

```
augment(m) |>  
  select(log_les, .fitted, .resid)
```

```
  log_les .fitted .resid  
1 -1.85    -0.473 -1.38  
2  0.0275  -1.59   1.62  
3 -0.538   -1.20   0.659  
4 -0.627   -0.526 -0.101  
# ... with 442 more rows
```

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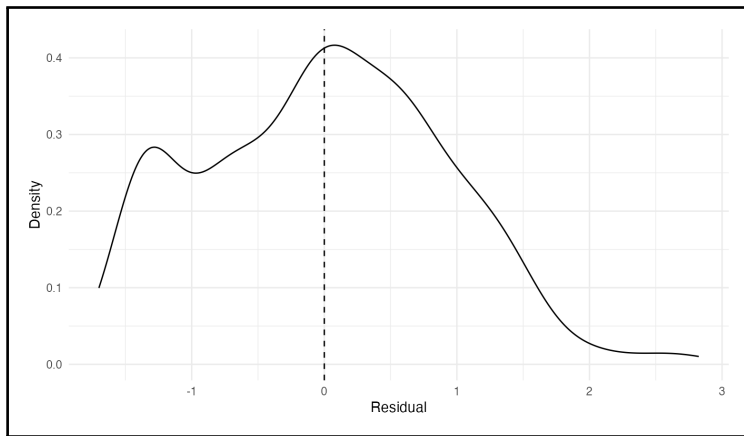
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# ... with 442 more rows
```

- ▶ **augment()** (from **broom**) adds the fitted value (**.fitted**) and residual (**.resid**) for every member

How Are the Errors Distributed?

```
augment(m) |>
  ggplot(aes(x = .resid)) +
  geom_density() +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = "Residual", y = "Density") +
  theme_minimal()
```

How Are the Errors Distributed?



Centered on zero: OLS guarantees the residuals average out. But some misses are large: a residual of ± 2 on the log scale means the actual LES is about 7 times higher or lower than predicted.

Who Does the Model Miss?

```
augment(m, data = house118) |>  
  select(name, bills, les, .fitted, .resid) |>  
  arrange(desc(.resid))
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```

	name	bills	les	.fitted	.resid
1	Granger, Kay	5	2.88	-1.77	2.82
2	Rogers, Mike	4	1.95	-1.98	2.65
3	Harris, Andy	2	0.917	-2.65	2.57
4	Graves, Sam	17	6.79	-0.581	2.50
#	... with 442 more rows				

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- ▶ Granger (Appropriations), Rogers (Armed Services), and Graves (Transportation) are **full committee chairs**; Harris chairs an Appropriations subcommittee (Cardinal)

Who Does the Model Miss?

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- ▶ Granger (Appropriations), Rogers (Armed Services), and Graves (Transportation) are **full committee chairs**; Harris chairs an Appropriations subcommittee (Cardinal)
- ▶ The errors aren't random: they point to something the model leaves out

Sum of Squared Residuals

- ▶ One number for total error: square each residual (so misses above and below don't cancel) and add them up: $RSS = \sum (y_i - \hat{y}_i)^2$

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```
augment(m) |>  
  summarise(rss = sum(.resid^2))  
  
glance(m) |>  
  select(deviance)
```

Sum of Squared Residuals

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augment(m) |>  
  summarise(rss = sum(.resid^2))
```

```
glance(m) |>  
  select(deviance)
```

```
      rss  
1  366.  
deviance  
1    366.
```

Sum of Squared Residuals

- One number for total error: square each residual (so misses above and below don't cancel) and add them up: $RSS = \sum (y_i - \hat{y}_i)^2$

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Problem: 366 is in squared logged-LES units, so we can't compare it across models with different outcomes.

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- Logged bills sponsored explains **36%** of the variation in logged LES.

Relationships with Categorical Variables

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Categorical Explanatory Variables

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- ▶ Our residuals pointed to committee chairs. So: **do committee chairs have higher effectiveness scores than other members?**
- ▶ With a categorical variable, we compare the outcome across groups instead of fitting a line

```
house118 <- house118 |>
  mutate(chair = if_else(chair == 1, "Committee chair", "Not a chair"),
         party = if_else(democrat == 1, "Democrat", "Republican"))

datasummary_crosstab(chair ~ party, data = house118)
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chair		Democrat	Republican	All
Committee chair	N	0	22	22
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Label the categories, then cross-tabulate. Every chair is a Republican: the majority party holds every gavel.

Comparing Group Means

```
house118 |>
  group_by(chair) |>
  summarise(
    mean = mean(les),
    sd = sd(les),
    freq = n()
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Comparing Group Means

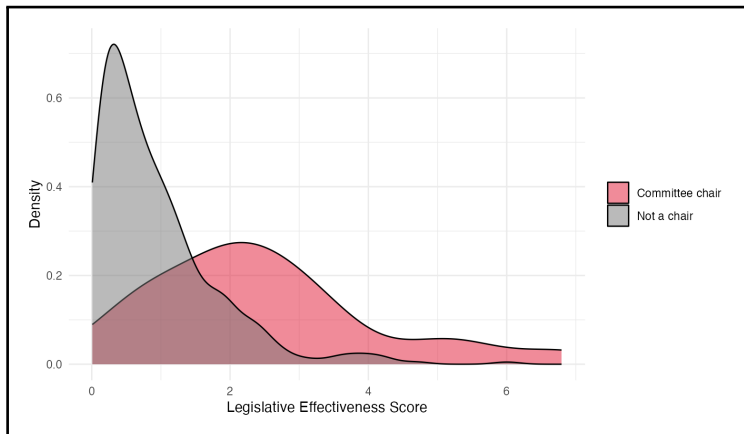
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- ▶ The average chair scores about **2.7 times** as high as the average non-chair
- ▶ Chairs also vary much more: the SD is nearly twice as large

```
ggplot(house118, aes(x = les, fill = chair)) +  
  geom_density(alpha = 0.5) +  
  scale_fill_manual(values = c("#E21833", "#767676")) +  
  labs(x = "Legislative Effectiveness Score", y = "Density", fill =  
    NULL) +  
  theme_minimal()
```

Comparing Distributions



Non-chairs pile up below 1; chairs are spread out and shifted to the right. Only 22 chairs, though, so their curve is based on far less data.

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 - ▷ With only 22 chairs, could a gap this size arise by chance? → hypothesis testing
 - ▷ Every chair is in the majority party, which controls the floor and committee agendas. Is it the gavel or the majority? → multiple regression
- ▶ The same confounding logic from last week's lab applies here: chairs and non-chairs differ in more ways than just the gavel

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- ▶ Residuals, RSS, and R^2 tell us how much the line leaves unexplained, and big residuals point to what's missing
- ▶ For categorical variables, compare group means and distributions

Center for Effective Lawmaking. 2025. "House Legislative Effectiveness Scores, 93rd–118th Congresses." <https://thelawmakers.org>.

Volden, Craig, and Alan E. Wiseman. 2014. *Legislative Effectiveness in the United States Congress: The Lawmakers*. New York: Cambridge University Press.

Questions?